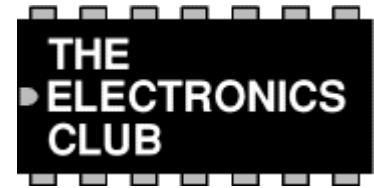


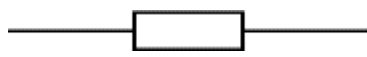
[Home](#) | [Capacitor](#) | [Connector](#) | [Diode](#) | [IC](#) | [Lamp](#) | [LED](#) | [Relay](#) | [Resistor](#) | [Switch](#) | [Transistor](#) | [Variable Resistor](#) | [Other](#)

Resistors



[Colour Code](#) | [Tolerance](#) | [Real Values \(E6 & E12 series\)](#) | [Power Rating](#)

Also see: [Resistance](#) | [Ohm's Law](#)

Example:  Circuit symbol: 

Function

Resistors restrict the flow of electric current, for example a resistor is placed in series with a light-emitting diode (LED) to limit the current passing through the LED.

Connecting and soldering

Resistors may be connected either way round. They are not damaged by heat when soldering.

Resistor values - the resistor colour code

Resistance is measured in ohms, the symbol for ohm is an omega Ω .
 1 Ω is quite small so resistor values are often given in k Ω and M Ω .
 1 k Ω = 1000 Ω 1 M Ω = 1000000 Ω .

Resistor values are normally shown using coloured bands.
 Each colour represents a number as shown in the table.

Most resistors have 4 bands:

- The **first band** gives the **first digit**.
- The **second band** gives the **second digit**.
- The **third band** indicates the **number of zeros**.
- *The fourth band is used to show the tolerance (precision) of the resistor, this may be ignored for almost all circuits but further details are given [below](#).*

The Resistor Colour Code	
Colour	Number
Black	0
Brown	1
Red	2
Orange	3
Yellow	4
Green	5
Blue	6
Violet	7
Grey	8
White	9



This resistor has red (2), violet (7), yellow (4 zeros) and gold bands.
 So its value is 270000 Ω = 270 k Ω .
 On circuit diagrams the Ω is usually omitted and the value is written 270K.

Find out how to make your own [Resistor Colour Code Calculator](#)

Small value resistors (less than 10 ohm)

The standard colour code cannot show values of less than 10 Ω . To show these small values two special colours are used for the **third band**: **gold** which means $\times 0.1$ and **silver** which means $\times 0.01$. The first and second bands represent the digits as normal.

For example:

red, **violet**, **gold** bands represent $27 \times 0.1 = 2.7 \Omega$

green, **blue**, **silver** bands represent $56 \times 0.01 = 0.56 \Omega$

Tolerance of resistors (fourth band of colour code)

The tolerance of a resistor is shown by the **fourth band** of the colour code. Tolerance is the **precision** of the resistor and it is given as a percentage. For example a 390 Ω resistor with a tolerance of $\pm 10\%$ will have a value within 10% of 390 Ω , between $390 - 39 = 351\Omega$ and $390 + 39 = 429\Omega$ (39 is 10% of 390).

A special colour code is used for the **fourth band** tolerance:

silver $\pm 10\%$, **gold** $\pm 5\%$, **red** $\pm 2\%$, **brown** $\pm 1\%$.

If no fourth band is shown the tolerance is $\pm 20\%$.

Tolerance may be ignored for almost all circuits because precise resistor values are rarely required.

Resistor shorthand

Resistor values are often written on circuit diagrams using a code system which avoids using a decimal point because it is easy to miss the small dot. Instead the letters R, K and M are used in place of the decimal point. To read the code: replace the letter with a decimal point, then multiply the value by 1000 if the letter was K, or 1000000 if the letter was M. The letter R means multiply by 1.

For example:

560R means 560 Ω

2K7 means 2.7 k Ω = 2700 Ω

39K means 39 k Ω

1M0 means 1.0 M Ω = 1000 k Ω

Real resistor values (the E6 and E12 series)

You may have noticed that resistors are not available with every possible value, for example 22k Ω and 47k Ω are readily available, but 25k Ω and 50k Ω are not!

Why is this? Imagine that you decided to make resistors every 10 Ω giving 10, 20, 30, 40, 50 and so on. That seems fine, but what happens when you reach 1000? It would be pointless to make 1000, 1010, 1020, 1030 and so on because for these values 10 is a very small difference, too small to be noticeable in most circuits. In fact it would be difficult to make resistors sufficiently accurate.

To produce a sensible range of resistor values you need to increase the size of the 'step' as the value increases. The standard resistor values are based on this idea and they form a series which follows the same pattern for every multiple of ten.

The E6 series (6 values for each multiple of ten, for resistors with 20% tolerance)

10, 15, 22, 33, 47, 68, ... then it continues 100, 150, 220, 330, 470, 680, 1000 etc.

Notice how the step size increases as the value increases. For this series the step (to the next value) is roughly half the value.

The E12 series (12 values for each multiple of ten, for resistors with 10% tolerance)

10, 12, 15, 18, 22, 27, 33, 39, 47, 56, 68, 82, ... then it continues 100, 120, 150 etc.

Notice how this is the E6 series with an extra value in the gaps.

The E12 series is the one most frequently used for resistors. It allows you to choose a value within 10% of the precise value you need. This is sufficiently accurate for almost all projects and it is sensible because most resistors are only accurate to $\pm 10\%$ (called their 'tolerance'). For example a resistor marked 390Ω could vary by $\pm 10\% \times 390\Omega = \pm 39\Omega$, so it could be any value between 351Ω and 429Ω .

Resistors in Series and Parallel

For information on resistors connected in series and parallel please see the [Resistance](#) page,

Power Ratings of Resistors

Electrical energy is converted to heat when current flows through a resistor. Usually the effect is negligible, but if the resistance is low (or the voltage across the resistor high) a large current may pass making the resistor become noticeably warm. The resistor must be able to withstand the heating effect and resistors have power ratings to show this.

Power ratings of resistors are rarely quoted in parts lists because for most circuits the standard power ratings of 0.25W or 0.5W are suitable. For the rare cases where a higher power is required it should be clearly specified in the parts list, these will be circuits using **low value resistors** (less than about 300Ω) or **high voltages** (more than 15V).

The power, P, developed in a resistor is given by:

$$P = I^2 \times R \quad \text{where: } P = \text{power developed in the resistor in watts (W)}$$

or

$$P = \frac{V^2}{R} \quad I = \text{current through the resistor in amps (A)}$$

R = resistance of the resistor in ohms (Ω)

V = voltage across the resistor in volts (V)



High power resistors
(5W top, 25W bottom)

Photographs © [Rapid Electronics](#)

Examples:

- A 470Ω resistor with 10V across it, needs a power rating $P = V^2/R = 10^2/470 = 0.21W$.

In this case a standard 0.25W resistor would be suitable.

- A 27Ω resistor with 10V across it, needs a power rating $P = V^2/R = 10^2/27 = 3.7W$.
A high power resistor with a rating of 5W would be suitable.

[Home](#) | [Capacitor](#) | [Connector](#) | [Diode](#) | [IC](#) | [Lamp](#) | [LED](#) | [Relay](#) | [Resistor](#) | [Switch](#) | [Transistor](#) | [Variable Resistor](#) | [Other](#)

- [Electronics Club Home Page](#)
- [Site Map](#)
- [Example Projects](#)
- [Construction of Projects](#)
- [Soldering Guide](#)
- [Study Electronics](#)
- [Electronic Components](#)
- [555 Timer](#)
- [Circuit Symbols](#)
- [Frequently Asked Questions](#)
- [Links to other Electronics sites](#)



© John Hewes 2007, The Electronics Club, www.kpsec.freeuk.com